

Assignment 1: Internet Explorer 9th Feb 2026

Client–Server Architecture and What Happens When You Open a Website

1. What is Client–Server Architecture?

Client–Server Architecture is a model where:

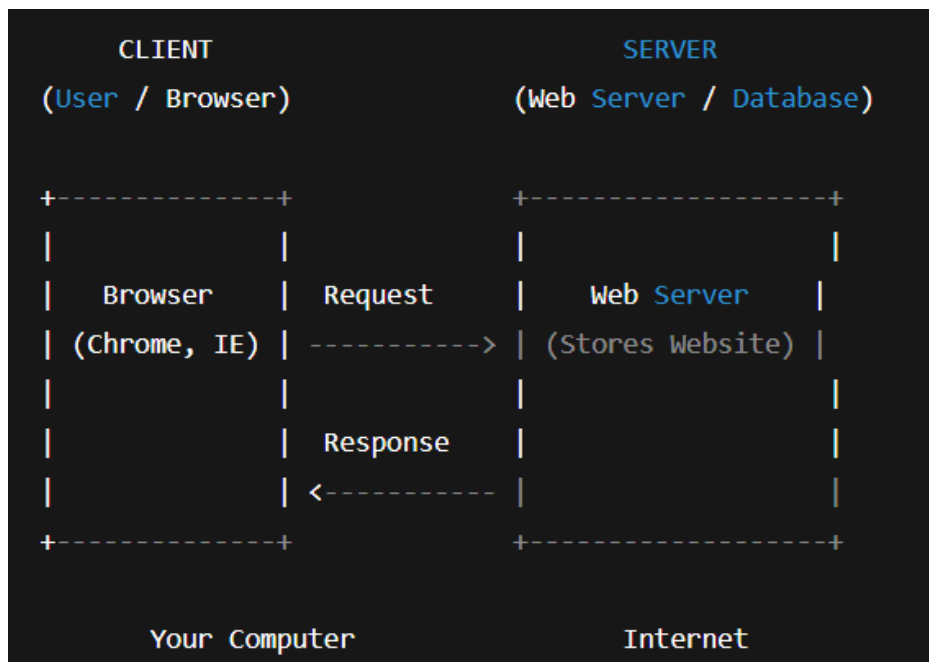
- **Client** → The device or browser that requests data
- **Server** → The computer that provides the requested data

Example:

When you open www.google.com, your browser is the **client**, and Google's computer is the **server**.

2. Diagram of Client–Server Architecture

You can draw this in your notebook or assignment like this:



You can also label:

- Client → Browser (Chrome, Edge, Firefox)
 - Server → Website hosting computer
 - Internet → Medium connecting them
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3. What Happens When You Open a Website (Step-by-Step)

Example: Opening www.youtube.com

Step 1: User enters URL

You type:

www.youtube.com

in the browser.

Browser acts as the **Client**.

Step 2: DNS Lookup

DNS (Domain Name System) converts:

www.youtube.com → IP Address (example: 142.250.183.206)

Because computers understand only IP addresses.

Step 3: Browser sends Request to Server

Browser sends an **HTTP Request** to the server.

Example:

GET / HTTP/1.1

This means:

"Hey server, send me the website."

Step 4: Server processes the request

Server:

- Receives request
 - Finds website files (HTML, CSS, JS)
 - Processes logic if needed
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Step 5: Server sends Response

Server sends **HTTP Response** back to browser:

Example:

HTML

CSS

JavaScript

Images

Step 6: Browser renders the website

Browser:

- Reads HTML
- Applies CSS (design)
- Executes JavaScript (functionality)

And shows the website on screen.

4. Real-Life Example Analogy

Client = You

Server = Restaurant kitchen

Process:

1. You order food → Request
 2. Kitchen prepares food → Processing
 3. Kitchen gives food → Response
 4. You eat → Result
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5. Components involved

- Client → Browser
 - Server → Web Server
 - Internet → Communication medium
 - DNS → Converts domain to IP
 - HTTP/HTTPS → Communication protocol
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6. Summary (Short Version – For Assignment)

Client–Server Architecture is a model where the client (browser) sends a request to the server, and the server responds with the requested data. When a user opens a website, the browser sends a request to the server using HTTP protocol. DNS converts the domain name into an IP address. The server processes the request and sends back website files such as HTML, CSS, and JavaScript. The browser then renders these files and displays the website to the user.